From Face Perception to Social Categorization: Multidimensional Scaling Approach to Ethnic Outgroup Homogeneity in Childhood

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"In 2023, Member States issued more than 3.7 million residence permits to non-EU citizens from more than 170 countries."

https://www.europarl.europa.eu/doceo/document/E-10-2025-000294-ASW_EN.html

"The study on fiscal impact of migration is a Science for Policy report by the Joint Research Centre, which provides evidence-based scientific support to the European policymaking process. [...]

The study also finds that removing obstacles to the full labour market integration of migrants can produce considerable positive fiscal benefits for hosting countries in the future."

https://www.europarl.europa.eu/doceo/document/E-10-2025-002052-ASW_EN.html

Categorization's effects

- Social categorization changes how people perceive, evaluate, remember, and treat one another (Amodio & Cikara, 2021; Levy et al., 2023; Rhodes & Baron, 2019)

Why is social categorization a relevant phenomenon in childhood?

- Early-emerging ingroup preferences documented in infants (e.g., Kelly et al., 2009; Woo et al., 2020)
- Visual specialization for ingroup faces (Quinn et al., 2016)

The Outgroup Homogeneity Effect (OHE; Quattrone & Jones, 1981)

- Outgroup faces appear more homogeneous to each other than ingroup faces

The Outgroup Homogeneity Effect (with Fantastic Mr.Fox)

In-group	Outgroup Beans	Outgroup Boggins	Outgroup Bunce
			

Minority-groups homogeneity effect (Tepper & Gilovich, 2025)

- In a multigroup scenario
- Majority members see different minority outgroups as MORE similar to each other

Minority-groups homogeneity effect (with Fantastic Mr.Fox)

The Multigroup Context

In children, the situation is slightly different:

- No formal position consolidated
- Multigroup paradigms are not employed, even though they are more ecologically valid
- **Gap:** We do not know how children categorize faces in a multigroup environment

Research Question:

- How is the multiethnic perceptual space structured in childhood? Does it support homogeneity theories?
- Does ethnic categorization change across development?

Pre-registered studies: Method & Experimental Design

Sample:

- Study 1: N=249 (moderated, controlled lab setting)
- Study 2: N=101 (unmoderated, online validation)
- Age range: 3 to 8 years

Multidimensional scaling (MDS)

Expanding the Scope

- 4-group context: Italian ingroup + 3 distinct ethnic outgroups
- Relational focus: Facial dyads rather than isolated face evaluation
- Continuous similarity judgments (0-3 scale) instead of binary choices

Face Similarity Task Protocol:

- Incomplete block design
- 66 random dyads presented sequentially

The Face Similarity Task

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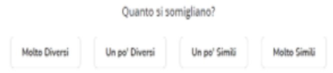
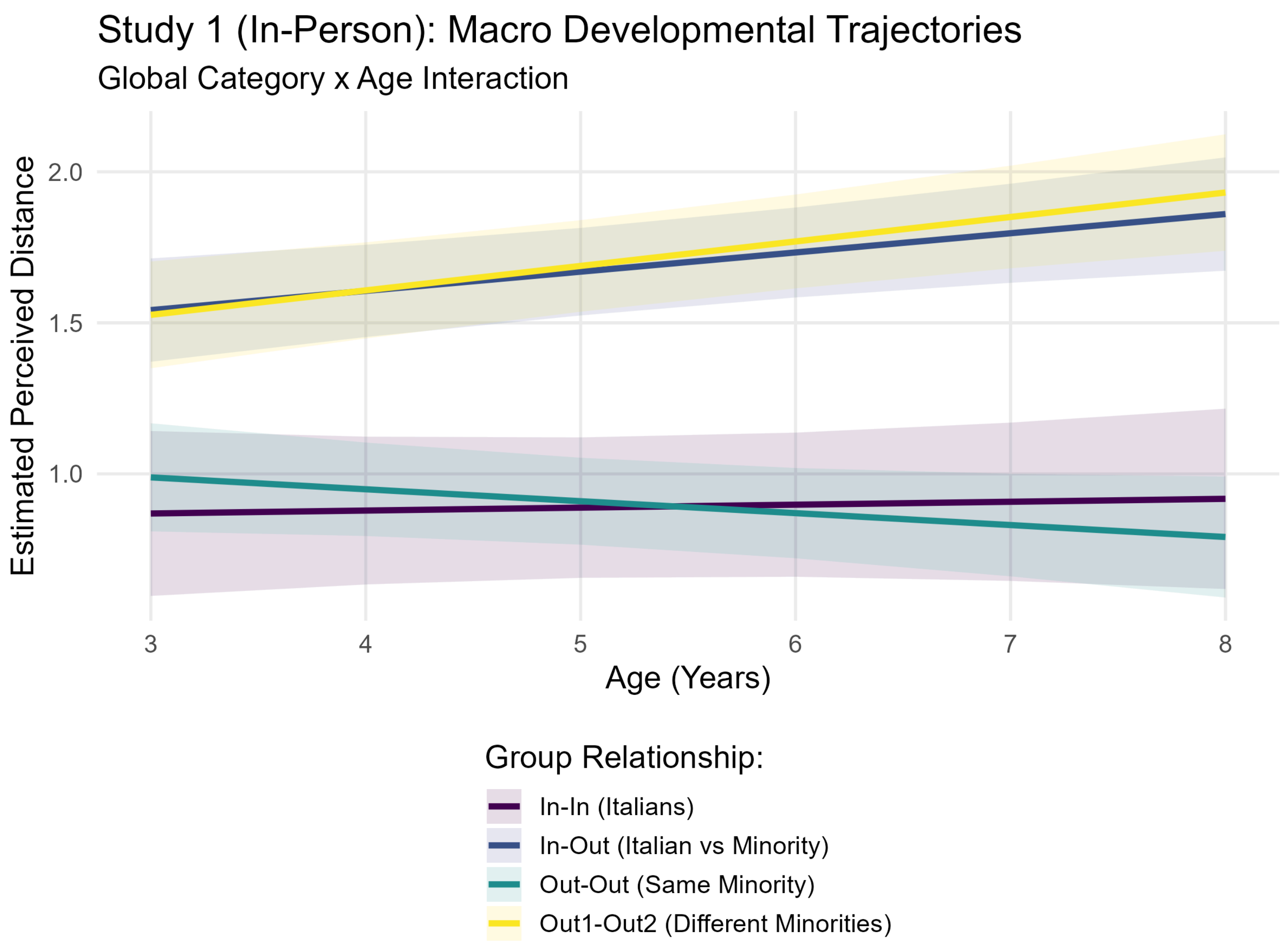
+		
800ms	1000ms	//

MDS

3D MDS visualization

- Children spontaneously visually categorize stimuli in ethnic groups

LMM - Developmental Trajectories



- Perceptual expertise in discriminating different social groups increases with age

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- Intergroup distinctions become more salient

Questions? 🫰